



Drugs for Acid-Related Disorders — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A 52-year-old man with a duodenal ulcer is started on omeprazole. At which step of gastric acid secretion does the active form of this drug act?

EASY

- A. Histamine release from ECL cells in the gastric fundus
- B. The H_2 receptor on the parietal cell membrane
- C. The H^+/K^+ ATPase (proton pump) of the parietal cell
- D. Gastrin release from G cells of the gastric antrum
- E. The M_3 muscarinic receptor on the parietal cell membrane

2. A student is told that ranitidine lowers acid output by acting on parietal cells. Which intracellular change explains this?

EASY

- A. Reduced cAMP, since H_2 receptors act via adenylate cyclase
- B. Reduced IP_3/Ca^{2+} , since H_2 receptors act via phospholipase C
- C. Increased cAMP, since the drug stimulates adenylate cyclase
- D. Irreversible covalent inactivation of the H^+/K^+ ATPase pump
- E. Blocked gastrin binding to CCK_2 receptors on the parietal cell

3. A 40-year-old man taking $Mg(OH)_2$ alone for heartburn has frequent loose stools. Which antacid could be added to counterbalance this bowel effect?

EASY

- A. Sodium bicarbonate, which is absorbed and slows gut motility
- B. Misoprostol, which reduces intestinal secretion and motility
- C. Magnesium hydroxide at a higher dose, which firms the stool
- D. Pantoprazole, which neutralises acid and firms the stool
- E. Aluminium hydroxide, whose Al^{3+} ions cause constipation

4. A 58-year-old woman with rheumatoid arthritis takes an NSAID daily and is at high risk of gastric ulcer. Which synthetic PGE_1 derivative can be given, with a PPI, to prevent NSAID-induced gastric ulcers?

EASY

- A. Sucralfate
- B. Misoprostol
- C. Pirenzepine
- D. Sodium bicarbonate
- E. Bismuth subsalicylate

5. A 40-year-old man with a duodenal ulcer tests positive for *H. pylori* and is given triple therapy for 2 weeks. Which regimen is triple therapy?

MEDIUM

- A. Bismuth subsalicylate + metronidazole + tetracycline + a PPI
- B. A PPI plus an H_2 blocker plus sucralfate for 2 weeks
- C. Amoxicillin + metronidazole + tetracycline, with no PPI
- D. A PPI with amoxicillin (or metronidazole) plus clarithromycin
- E. Misoprostol + clarithromycin + bismuth subsalicylate

6. A 45-year-old woman with GERD takes omeprazole right after breakfast and still has symptoms. What advice best improves control, and why?

MEDIUM

- A. Take it with a fatty meal, because food doubles its bioavailability and speeds onset
- B. Take it ~1 hour before a meal, so peak levels coincide with maximal pump activity
- C. Take it at bedtime only, because the proton pump is active only during sleep
- D. Split it into four daily doses, because a 1.5-hour half-life lets acid return
- E. Take it together with an antacid, which is needed to activate the prodrug

7. A patient asks why one omeprazole dose a day controls acid for a full day when its serum half-life is only about 1.5 hours. What is the best explanation?

MEDIUM

- A. Its active metabolite accumulates in the kidney and is then released slowly over 24 hours
- B. It is a competitive antagonist that dissociates from the H₂ receptor very slowly
- C. It stimulates mucus and bicarbonate secretion for a full 24 hours
- D. It forms a viscous barrier over the ulcer crater that lasts a full day
- E. It inactivates the pump irreversibly; ~18 hours are needed to resynthesise the enzyme

8. A 68-year-old man who has taken a PPI for 8 years has a subnormal serum vitamin B12. Which explanation fits the lecture?

MEDIUM

- A. Acid releases B12 from food, and long-term PPI therapy lowers acid
- B. PPIs bind B12 in the gut lumen and prevent it being absorbed in the ileum
- C. PPIs permanently destroy the parietal cells that secrete intrinsic factor
- D. PPIs induce hepatic CYP450 enzymes that break down circulating B12
- E. PPIs alkalinise the urine, which markedly increases urinary B12 excretion

9. A 35-year-old man has recurrent severe ulcers and marked acid hypersecretion; a gastrin-secreting tumour is found (Zollinger-Ellison syndrome). Which drug class is used for long-term treatment, being superior to H₂ blockers at suppressing acid?

MEDIUM

- A. Antacids taken after meals
- B. Misoprostol
- C. Proton pump inhibitors
- D. Sucralfate
- E. Bismuth subsalicylate

10. A 62-year-old man stable on warfarin and phenytoin starts an H₂ blocker for a gastric ulcer and soon shows warfarin and phenytoin toxicity. Which H₂ blocker is most likely responsible?

MEDIUM

- A. Ranitidine
- B. Famotidine
- C. Nizatidine
- D. Cimetidine
- E. Sucralfate

11. An 82-year-old woman with moderate renal impairment needs an H₂ blocker for a duodenal ulcer. What is the correct approach?

MEDIUM

- A. No adjustment, because H₂ blockers are cleared entirely by the liver
- B. Reduce the dose, because clearance falls in renal insufficiency and in the elderly
- C. Increase the dose, because first-pass metabolism halves bioavailability in the elderly
- D. Avoid the whole class, because H₂ blockers cause permanent loss of pump function
- E. Use nizatidine at full dose, because it is not eliminated by the kidney at all

12. A 29-year-old woman on long-term NSAIDs is started on misoprostol to prevent ulcers. Which pair of statements is correct?

MEDIUM

- A. Check renal function (risk of aluminium toxicity); constipation is the main adverse effect
- B. Monitor phenytoin levels (CYP450 inhibition); gynaecomastia is the main adverse effect
- C. Monitor serum B12 (from reduced acid output); muscle pain is the main adverse effect
- D. Monitor serum phosphate (phosphate binding); black stools are the main adverse effect
- E. Exclude pregnancy (uterine contractions); dose-related diarrhoea is the main adverse effect

13. A man with a duodenal ulcer takes sucralfate, then calcium carbonate 10 minutes later for heartburn. Why is this combination undesirable?

MEDIUM

- A. Sucralfate needs an acidic pH for activation, which antacids remove
- B. Antacids stimulate the proton pump, which dissolves the sucralfate barrier
- C. Sucralfate binds only intact mucosa, never ulcers, once the pH is raised
- D. Calcium carbonate induces CYP450 and speeds up sucralfate metabolism
- E. Sucralfate binds the calcium carbonate, so no antacid effect occurs

14. A 30-year-old woman with a urinary tract infection takes ciprofloxacin (a quinolone) together with an Mg/Al antacid, and the infection responds poorly. Which explanation is correct?

MEDIUM

- A. The antacid induces hepatic CYP450 enzymes and speeds up ciprofloxacin metabolism
- B. The antacid stimulates the proton pump, so ciprofloxacin is degraded in the stomach
- C. Mg^{2+} and Al^{3+} chelate the quinolone into insoluble complexes, reducing absorption
- D. The antacid competes with ciprofloxacin for H_2 receptors and blocks its action
- E. Antacid-induced hypophosphatemia makes the urinary bacteria drug-resistant

15. In an experiment, famotidine blocks parietal-cell H_2 receptors, yet vagal stimulation still produces some acid secretion. Which statement best explains this residual secretion?

HARD

- A. Vagal ACh stimulates the pump only through H_2 receptors, so blockade cannot change secretion
- B. Famotidine is an irreversible inhibitor and cannot displace histamine from its receptor
- C. The proton pump is activated only by cAMP generated by M_3 receptors on parietal cells
- D. ACh (M_3) and gastrin (CCK_2) also stimulate the pump directly via IP_3/Ca^{2+} , bypassing H_2
- E. ECL cells stop releasing histamine, but G cells release more gastrin, which acts through H_2

16. Misoprostol lowers histamine-stimulated cAMP in parietal cells, yet a 24-hour pH study shows less overall acid suppression than with an H_2 blocker. What is the main reason?

HARD

- A. Its duration of action is short, about 3 hours
- B. It raises cAMP, which opposes its acid-lowering effect
- C. It acts only on gastrin release and never on parietal cells
- D. It is eliminated only by the kidney and needs dose reduction
- E. It is a natural prostaglandin with a very long half-life that accumulates

17. A 50-year-old man completing quadruple therapy for an *H. pylori* duodenal ulcer has a black tongue and black stools, with no abdominal pain, dizziness or other alarming features. Which statement is most accurate?

HARD

- A. They signal tetracycline-induced hepatotoxicity, so the regimen must be stopped at once
- B. They indicate metronidazole-induced ulcer bleeding, so misoprostol should be added
- C. They are typical of PPI-induced vitamin B12 deficiency and need no action
- D. They mean sucralfate has stained the mucosa, so it should be stopped
- E. Both are expected effects of bismuth subsalicylate, which also coats the ulcer base

18. A 62-year-old man with chronic renal failure has taken an aluminium hydroxide antacid daily for months and now has bone pain and confusion. Which pair of mechanisms best explains this?

HARD

- A. Systemic alkalosis from absorbed bicarbonate, plus a sodium load that worsens oedema and heart failure
- B. Phosphate binding causes hypophosphatemia and osteomalacia; poor renal excretion lets aluminium accumulate
- C. Magnesium accumulation, causing hyporeflexia, hypotension and eventually cardiac arrest in renal failure
- D. Hyperphosphatemia from phosphate release, plus excessive calcium loss in the urine and a raised urinary pH
- E. CO_2 liberation in the stomach, causing distension, belching and a risk of ulcer perforation

19. A 66-year-old man with heart failure takes sodium bicarbonate repeatedly for heartburn and develops worsening ankle oedema, belching and epigastric discomfort. Which two mechanisms explain these findings?

HARD

- A. Al^{3+} ions delay gastric emptying, and phosphate binding causes hypophosphatemia and osteomalacia
- B. Mg^{2+} accumulation lowers blood pressure, and osmotic diarrhoea causes loss of body fluid
- C. Sodium load worsens oedema/heart failure; CO_2 formed in the stomach causes distension and belching
- D. Irreversible pump blockade causes fluid retention, and B12 deficiency causes belching
- E. Urine alkalisation retains sodium, and chelation of gastric acid causes belching

20. A patient with Zollinger-Ellison syndrome stays symptomatic on standard therapy, so an antimuscarinic is added. He later has palpitations, dry mouth, constipation and urinary hesitancy. Which statement about this class is correct?

HARD

- A. Pirenzepine blocks M_1 , M_2 and M_3 , while dicyclomine is selective for M_1 receptors
- B. They are first-line agents for peptic ulcer disease and are preferred over PPIs
- C. They lower acid by blocking H_2 receptors, which also explains the dry mouth
- D. Pirenzepine is a selective M_1 blocker, dicyclomine blocks M_1 - M_3 , and side effects limit both
- E. Their adverse effects come from histamine blockade rather than from muscarinic receptor blockade

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	PPIs bind the H ⁺ /K ⁺ ATPase irreversibly — the final step in acid secretion.
2	A	EASY	H ₂ antagonists compete reversibly at H ₂ receptors, lowering cAMP and so acid secretion.
3	E	EASY	Mg(OH) ₂ causes diarrhoea, Al(OH) ₃ causes constipation; combined, they may cancel each other's bowel effects.
4	B	EASY	Misoprostol (PGE ₁ analogue) with a PPI prevents NSAID ulcers; sucralfate does not.
5	D	MEDIUM	Triple therapy = PPI + (metronidazole or amoxicillin) + clarithromycin; the bismuth-based four-drug regimen is quadruple therapy.
6	B	MEDIUM	Food cuts PPI bioavailability by 50%; give ~1 h before a meal so peak levels meet maximal pump activity.
7	E	MEDIUM	Irreversible pump inactivation: ~18 h needed for resynthesis, so effect lasts up to 24 h despite t _{1/2} ≈ 1.5 h.
8	A	MEDIUM	Acid releases B12 from food; long-term PPI use can therefore lower B12 levels.
9	C	MEDIUM	PPIs are superior to H ₂ antagonists for acid suppression and are used long-term in Zollinger–Ellison syndrome.
10	D	MEDIUM	Cimetidine inhibits CYP450, potentiating warfarin, phenytoin, diazepam, quinidine, carbamazepine, theophylline and imipramine.
11	B	MEDIUM	H ₂ blockers need dose reduction in moderate–severe renal insufficiency and in the elderly; nizatidine is mainly renally eliminated.
12	E	MEDIUM	Misoprostol causes uterine contractions (contraindicated in pregnancy); dose-related diarrhoea and nausea limit its use.
13	A	MEDIUM	Sucralfate requires acid for activation, so it should not be given with H ₂ antagonists or antacids.
14	C	MEDIUM	Ions in antacids chelate drugs — e.g. quinolones with Ca ²⁺ , Mg ²⁺ or Al ³⁺ — forming insoluble complexes.
15	D	HARD	ACh and gastrin also stimulate the pump directly through a Ca ²⁺ -dependent pathway, separate from H ₂ /cAMP.
16	A	HARD	Misoprostol's duration is only ~3 h, so 24-hour acid inhibition is less than with H ₂ blockers.
17	E	HARD	Bismuth subsalicylate blackens tongue and faeces; it also has antimicrobial action, coats the ulcer base and inhibits pepsin.
18	B	HARD	Al ³⁺ binds phosphate (hypophosphatemia → osteomalacia); in renal failure Al ³⁺ is not excreted → encephalopathy, osteoporosis.
19	C	HARD	NaHCO ₃ is absorbed: sodium load (worsens oedema/CHF) and alkalosis; CO ₂ in the stomach → distension, belching, perforation risk.
20	D	HARD	Antimuscarinics (pirenzepine M ₁ ; dicyclomine M ₁ /M ₂ /M ₃) are reserved for refractory PUD/ZES; arrhythmias, dry mouth, constipation, urinary retention limit use.

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Begard O. Berzinji lecture slides, SGL 1 (full deck).

Practice questions — cross-check every answer against the lecture and summary.